

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

EE 19-209 -0405 DIGITAL ELECTRONICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Understand number systems and codes and fundamentals of Boolean algebra.
 CO2: Design various types of registers and counters using latches and flip-flops.
 CO3: Design and realise logical circuits for mathematical operations.
 CO4: Understand various memories and logic families and its characteristics.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) (i) Convert 106.15 to binary.	3	L2	1	1
(ii) Subtract 14 from 45 using 8 bit 2's complement.	3	L1	1	1
(b) State and explain De-Morgan's theorem.	3	L1	2	1
(c) Explain the working of a 2 bit SIPO shift register.	3	L1,L2	2	1
(d) Convert J K Flip Flop to T Flip Flop.	3	L2	3	1
(e) Compare PROM, PLA and PAL.	3	L1,L2	3	1
(f) Obtain the logic implementation 8 × 3 ROM.	3	L1	4	1,2
(g) Explain the operation of a CMOS inverter.	3	L1,L2	4	1,2
(h) Differentiate between RTL and DTL.	3			

PART B

(4 × 12 = 48)

II.	Obtain the minimal expression using Quine Mc Clusky method $F(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$.	12	L2,L3	1	1
OR					
III.	Simplify the function using K map and realise using only NAND gates $F(A, B, C, D) = \sum m(0, 2, 6, 10, 12, 13) + d(3, 4, 5, 14, 15)$.	12	L2,L3	1	1
IV. (a)	Realise the function $F(A, B, C, D) = \sum m(1, 2, 4, 7, 8, 10, 11, 12)$ using 8:1 multiplexer.	6	L2,L3	2	1
(b)	Draw the logic diagram of a 4 stage look ahead carry adder.	6	L1,L2	2	1
OR					
V. (a)	Design and implement a full subtractor using 2 half subtractors.	6	L3	2	1
(b)	Explain the working of astable multivibrators using gates.	6	L1,L2	2	1

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		Marks	BL	CO	PO
VI.	Design a mod-8 asynchronous counter using J-K flip flops. Draw the timing diagram.	12	L1,L2	3	1
OR					
VII. (a)	Show how to implement a full adder using PAL Circuit.	6	L1,L2	3	1
(b)	Implement $F1 = BC + ABC$ and $F2 = C + B + ABC$ using PAL and PLA.	6	L3	3	1
VIII. (a)	Draw the circuit of TTL tristate inverter and explain its operation.	8	L1,L2	4	1
(b)	Explain noise margin.	4	L1,L2	4	1
OR					
IX. (a)	Draw and explain the circuits of	7	L1,L2	4	1
	(i) CMOS to TTL interface				
	(ii) TTL to CMOS interface.				
(b)	Define the following terms	5	L1,L2	4	1,2
	(i) fan in				
	(ii) fan out				
	(iii) propagation delay.				

Blooms's Taxonomy Level's

L1 - 34%, L2 - 46%, L3 - 20%.
